

Laksh Goyal

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EDUCATION

University of California San Diego San Diego, CA

B.S. in Computer Science, minor in Cognitive Science and Business-Economics

Sep 2024 - Jun 2028

- **Relevant Coursework:** Machine Learning; Probability & Statistics; Operating Systems Principles; Software Engineering; Systems Programming; Computer Architecture; Linear Algebra; Algorithms & Data Structures

EXPERIENCE

Inovacity San Diego, CA

Applied Machine Learning Intern

Mar 2026 - Present

- Building Python/PyTorch ML pipelines for 3D wound analysis using MedSAM2, DINOv2, and MONAI
- Generating synthetic wound datasets with ground-truth depth maps and tissue labels for pre-clinical model training
- Fine-tuning depth estimation and segmentation models in PyTorch to improve wound accuracy and model confidence
- Developing MAE, RMSE, and IoU validation workflows on 50+ samples for pre-clinical model evaluation

Qualcomm Institute - Engineers For Exploration San Diego, CA

Software Engineer (Machine Learning Systems)

Jan 2026 - Present

Research Engineer (Computer Vision)

Oct 2025 - Jan 2026

- Built Python ingestion and indexing pipelines for multi-terabyte acoustic datasets for CNN and EfficientNet training
- Increased usable audio segments by 15% by automating audio QC and corrupted-file filtering in large-scale workflows
- Reduced preprocessing time by 10% through faster batch throughput, metadata indexing, and NAS file handling
- Analyzing focal-to-soundscape domain shift in bioacoustics using STFT, MFCC, PANNs, and WST representations
- Evaluating robustness with accuracy, F1, and distribution-shift metrics to identify the best-generalizing features
- Improved boundary accuracy by 9% by adding a boundary-aware IoU metric for mangrove canopy segmentation
- Refactored evaluation into a metric registry, decoupling metrics from the training loop to enable boundary-aware IoU

Xyntopia - Taskyon San Diego, CA

LLM Systems Engineer Intern

Jun 2025 - Oct 2025

- Built a 30+ stage agentic task orchestration framework in Python powering a VS Code chat extension
- Built a token-based E2E encrypted proxy server routing LLM and third-party API requests
- Implemented server-side token verification with automated credit reporting to a Supabase payment backend

- Reduced task retries by 35% using validation checks and state-aware partial retries instead of rerunning full tasks

Computer Science & Engineering Society @ UCSD San Diego, CA

iOS/Android Developer

Jun 2025 - Mar 2026

- Improved skin-lesion classification by 7% by fine-tuning MedGemma-1.5-4B-IT on a 25K+ labeled image dataset
- Trained a CNN on HAM10000 dataset to classify lesions with high accuracy using augmentation and preprocessing
- Built a cross-platform mobile app for real-time skin lesion detection and diagnostic feedback, tested on 200+ images
- Deployed a lightweight TFLite/Core ML model optimized for offline use on iOS/Android devices

Healthlink@UCSD San Diego, CA

Web & Software Developer

Jun 2025 - Present

- Led migration of website components from Webflow to a React-based system, reducing page load time by 20%
- Built a membership tracker using React and Supabase to log events and update membership status for 100+ members
- Coordinated development tasks and feature implementation across the team to support a more scalable platform

PROJECTS

Argus - Agentic Training Monitor

PyTorch, FastAPI, Anthropic API, React, Supabase, Docker, AWS EC2

- Built an autonomous ML training agent (claude-sonnet-4-5) polling live JSONL every 10s to self-correct training
- Dispatched up to 10 tool calls/invoke via a 12-rule system prompt, resolving 49% of detected anomalies autonomously
- Engineered a SPC detector across 4 failure modes (loss spike, grad explosion, plateau, overfitting) via z-score/CUSUM
- Architected a 4-service Docker Compose stack (PyTorch trainer, LLM agent, FastAPI, React) on 3 Supabase tables
- Deployed on AWS EC2 with a React dashboard: 3 live Recharts plots, 5s polling, dual JSONL/Supabase logging

AgriShield - Wildfire Risk & Firebreak Optimizer

Python, rasterio, GeoPandas, NumPy, SciPy, Pydantic, ELMFIRE, JavaScript, Canvas API

- Built a 7-stage geospatial preprocessing pipeline (4.6K lines) converting farm boundary polygons into 11 ELMFIRE-ready rasters
- Aligned all rasters to a canonical 30m EPSG:5070 grid using `reproject_match` with nearest-neighbor and bilinear resampling
- Integrated LANDFIRE (LFPS) and USGS 3DEP APIs to fetch fuel models, canopy properties, and elevation data with SHA-256-keyed caching
- Ran 8-bearing fire-spread ensemble simulations via ELMFIRE, modeling 6-hour burns at 30s timesteps from ignition points
- Scored 24 candidate firebreak layouts using a 6-weight composite function prioritizing protected-area safety and risk reduction

- Shipped a no-build HTML5 + Canvas API web UI (2.5K lines JS) with interactive polygon drawing and live simulation polling
- Achieved 92 passing tests with mypy strict mode (100% type coverage), ruff strict linting, and full Pydantic v2 config validation

Project Liftoff - Startup Incubator Management System

[React](#), [Supabase](#), [PostgreSQL](#), [Resend](#), [Railway](#)

- Built a full-stack platform with 6-tier RBAC enforced via Supabase Row-Level Security across 13 relational tables
- Replaced Google Form intake with a multi-stage application pipeline for cohort review, onboarding 100+ users
- Designed an event and points engine tracking attendance codes, per-event values, and cumulative member scores
- Integrated Resend API for tokenized transactional confirmation emails on event registration; deployed on Railway

UrbanSound8K Sound Classification

[Python](#), [LightGBM](#), [Kymatio](#), [librosa](#), [Optuna](#), [scikit-learn](#), [torchaudio](#)

- Built a CNN-free audio classification pipeline on UrbanSound8K (8,732 clips, 10 classes), achieving 72.9% top-1 accuracy, 0.737 macro F1, and 86.8% top-2 accuracy with LightGBM across strict 10-fold CV
- Engineered a 487-dim feature vector combining MFCCs, spectral features, ZCR, RMS, chroma, tonnetz, and Wavelet Scattering Transform coefficients (Kymatio, J=8, Q=12)
- Benchmarked 12 model-feature combinations with leak-free Optuna hyperparameter search; achieved 0.51ms per-clip inference latency
- Implemented domain shift robustness experiment with noise injection, EQ coloration, and synthetic reverb, quantifying accuracy degradation (80% to 66.5% under reverb)

Helmet Identification Model

[Azure ML](#), [Blob Storage](#), [OpenCV](#), [Python](#)

- Built a multi-stage computer vision pipeline detecting motorbike riders and classifying helmet usage using ResNet-50
- Trained on 5K+ street images, achieving 94% accuracy and 0.92 mAP to support traffic law enforcement
- Optimized for real-time deployment, achieving <50ms GPU inference while balancing accuracy-latency tradeoffs
- Published a research paper through the Lumiere Research Program, ranked Top 10% of ~200 submissions

PipeScoutX

[Python](#), [Flask](#), [React](#), [PostgreSQL](#), [Docker](#), [Arduino](#), [ESP32](#), [OpenCV](#)

- Prototyped a pipe-inspecting bot with object detection to detect pipeline cracks and prevent oil/gas leaks
- Integrated sensors using C, Arduino, and ESP32; applied OpenCV for real-time crack detection and localization
- Tested on 100+ meters of pipeline with 90% detection accuracy; placed 2nd at World Robotic Olympiad India

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, Flutter

Frameworks/Libraries: PyTorch, FastAPI, MONAI, React, Flask, OpenCV, NumPy, pandas, Matplotlib, Open3D, TensorFlow, scikit-learn, LightGBM, Kymatio, librosa

Developer Tools: Git, Docker, AWS, Supabase, PostgreSQL, Azure ML, Blob Storage, Railway, Resend, Arduino IDE, ESP32, VS Code